

WIRESHARK LAB ACTIVITIES FOR BEGINNERS

ACTIVITY 1: CAPTURING BASIC NETWORK TRAFFIC

OBJECTIVE: LEARN HOW TO START AND STOP PACKET CAPTURE IN WIRESHARK.

INSTRUCTIONS:

- **OPEN WIRESHARK AND SELECT YOUR ACTIVE NETWORK INTERFACE.**
- **START PACKET CAPTURE BY CLICKING THE SHARK FIN ICON.**
- **OPEN A BROWSER AND VISIT ANY WEBSITE (E.G., [HTTPS://EXAMPLE.COM](https://example.com)).**
- **STOP THE CAPTURE AFTER 1 MINUTE.**
- **SAVE THE CAPTURE FILE AS BASIC_CAPTURE.PCAPNG.**

GUIDED QUESTIONS:

- 1. HOW MANY PACKETS WERE CAPTURED?**
- 2. WHAT IS YOUR COMPUTER'S IP ADDRESS?**
- 3. WHAT IS THE MOST COMMON PROTOCOL CAPTURED?**

ACTIVITY 2: EXPLORING PACKET STRUCTURE

OBJECTIVE: UNDERSTAND HOW WIRESHARK DISPLAYS PACKET LAYERS.

INSTRUCTIONS:

- **OPEN YOUR SAVED CAPTURE FROM ACTIVITY 1.**
- **CLICK ON ANY HTTP OR TCP PACKET.**
- **EXPAND THE FOLLOWING SECTIONS:**
 - **- FRAME (PACKET DETAILS)**
 - **- ETHERNET II**
 - **- INTERNET PROTOCOL VERSION 4**
 - **- TRANSMISSION CONTROL PROTOCOL**

GUIDED QUESTIONS:

- 4. WHAT IS THE LENGTH OF THE SELECTED PACKET?**
- 5. WHAT IS THE SOURCE AND DESTINATION IP?**
- 6. WHAT PORT IS BEING USED?**

ACTIVITY 3: USING DISPLAY FILTERS

OBJECTIVE: LEARN TO FILTER PACKETS BY PROTOCOL AND IP.

INSTRUCTIONS:

- **START A NEW CAPTURE.**

- VISIT MULTIPLE WEBSITES AND STOP THE CAPTURE.
- USE THE FOLLOWING FILTERS ONE AT A TIME:
- - HTTP
- - TCP
- - IP.ADDR == YOUR_COMPUTER_IP
- - ICMP (REQUIRES PING TEST)
- - DNS

GUIDED QUESTIONS:

7. HOW MANY PACKETS ARE DISPLAYED FOR EACH FILTER?
8. WHAT DOMAIN WAS QUERIED IN THE DNS PACKET?
9. WHAT IS THE RESPONSE CODE IN ONE HTTP PACKET?

ACTIVITY 4: DNS ANALYSIS

OBJECTIVE: UNDERSTAND HOW DNS REQUESTS WORK IN NETWORK COMMUNICATION.

INSTRUCTIONS:

- OPEN A TERMINAL AND RUN: PING GOOGLE.COM
- CAPTURE PACKETS WHILE PINGING.
- STOP CAPTURE AND FILTER WITH DNS.

GUIDED QUESTIONS:

10. WHAT IS THE IP ADDRESS RESOLVED FROM GOOGLE.COM?
11. HOW MANY DNS PACKETS WERE CAPTURED?
12. WAS THE DNS QUERY SUCCESSFUL?

ACTIVITY 5: FOLLOWING TCP STREAMS

OBJECTIVE: ANALYZE COMMUNICATION BETWEEN CLIENT AND SERVER.

INSTRUCTIONS:

- START CAPTURING PACKETS.
- VISIT A WEBSITE THAT USES HTTP (NOT HTTPS) LIKE HTTP://NEVERSSL.COM.
- STOP THE CAPTURE.
- LOCATE A TCP PACKET, RIGHT-CLICK → FOLLOW → TCP STREAM.

GUIDED QUESTIONS:

13. WHAT IS THE CONTENT OF THE HTTP REQUEST?
14. WHAT SERVER RESPONDED TO THE REQUEST?
15. DESCRIBE THE FLOW OF COMMUNICATION BETWEEN CLIENT AND SERVER.